Lilja Kiiski

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Education

Purdue University West Lafayette, IN

B.S. in Electrical Engineering, Minor in Mathematics; GPA: 3.81/4.0 *Aug. 2025 – May 2029*

- **Relevant Coursework:** EE Fundamentals I & II, Digital Design, Advanced C Programming, E&M Physics, Circuits Lab I, Engineering Honors Systems Design, Differential Equations, Linear Algebra.
- **Upcoming (Spring 2027):** Signals & Systems, Microprocessors, Electromagnetics I, Circuits Lab II.

Technical Experience

CS Extended Reality (XR) Lab West Lafayette, IN

Undergraduate Researcher *Aug. 2026 – Present*

- Designed a custom haptic device using six solenoids, an ESP32, and an IMU to render 3D force-direction feedback, simulating electric-field forces for XR applications.
- Designed a two-layer KiCad PCB that drives six 1 A solenoids through TIP120 transistors, replacing breadboard wiring to fit a tennis-ball-sized enclosure.
- Researched and evaluated voice coil actuators, LRAs, and solenoids, and selected solenoids for 25x lower cost and simpler circuitry.

Purdue IEEE Autonomous Motorsports West Lafayette, IN

Electrical Team Member *Aug. 2025 – May 2026*

- Created the schematic for an 84 V-to-19 V/12 V DC-DC converter PCB that powers the autonomous kart's lidar system, working from system requirements using KiCad and TI WEBENCH Power Designer.
- Validated the core converter circuitry in LTspice simulation under mentor guidance.

Technical Projects

4-Bit Carry-Save Adder | *SystemVerilog, Breadboarding, Analog Discovery 2, Oscilloscope*

- Built a 4-bit carry-save adder on a breadboard that sums three 4-bit inputs using eight full adders made from 10 logic ICs, with push-button inputs and LED outputs.
- Modeled the design in SystemVerilog and verified it with a testbench covering all 4,096 input combinations.
- Measured carry propagation delay on an oscilloscope at 50% thresholds: 4.22 μs rising and 1.04 μs falling.

Honeycomb Game Console | *C++, Arduino, Onshape, 3D Printing, Fritzing*

- Built a handheld game console around an Arduino Mega with a 3.5-inch TFT LCD, a joystick for aiming, and a limit switch as the fire button.
- Programmed an asteroid-shooter game in C++ that tracks hits, reaching a high score of 83 in testing.

Leadership & Teaching

Purdue University College of Engineering, Honors West Lafayette, IN

Undergraduate Teaching Assistant *Aug. 2026 – Present*

- Mentored 50+ students on labs, projects, and homework in MATLAB, Python, physics, and engineering design.
- Designed and led review sessions to help students prepare for midterms and exams.

Monta Vista Robotics Team Cupertino, CA

President, Computer Vision Lead *Oct. 2021 – Jun. 2025*

- Led a 150-member team through the design, build, and testing of a competitive FRC robot.
- Directed the software and electrical divisions in building the team's first AprilTag-based localization and alignment system, running a Java vision pipeline on Linux and Raspberry Pi for accurate robot self-localization on the field.
- Managed the team's schedule, budget, community outreach, and mentor/sponsor relations.

Technical Skills

Programming & Tools: C, Java, Python, MATLAB, SystemVerilog, Linux, Bash, Git, SSH
Hardware & Design: KiCad, LTspice, TI WEBENCH, ESP32, Arduino, Raspberry Pi, Onshape, 3D Printing
Lab Equipment: Oscilloscope, Digital Multimeter, Analog Discovery 2, Soldering, Breadboarding
Languages: English (Native), Finnish (Native), Spanish (Elementary)